



Chapter # 15

Pharmacological Drugs



Student Learning Outcomes

After studying this chapter, students will be able to:

- [B-12-T-01] Explain the drug discovery and development process.
- [B-12-T-02] Define 4 classes of antibiotics (Penicillins, Tetracyclins, Fluriquinolones and Sulfonamides) and describe their mode of action.
- [B-12-T-04] Define antivirals and antiretrovirals.
- [B-12-T-05] Describe advantages of monoclonal antibodies enjoy compared to other drug classes.

Drugs have played a central role in human health and disease management throughout history. In biology, the term *drug* refers to any chemical substance that, when taken into the body, alters its physiological processes. Some drugs are used for therapeutic purposes to prevent, treat, or cure diseases, while others may be misused, leading to harmful effects on health.

Pharmacological drugs are specifically those substances that act on the body's biological systems, particularly the nervous system, to produce characteristic effects. They may act by stimulating or depressing the activity of certain cells, blocking or enhancing the transmission of nerve impulses, or modifying biochemical reactions in tissues.

Understanding pharmacological drugs is essential for several reasons:

- **Medical Importance:** They form the foundation of modern medicine, enabling treatment of infections, mental disorders, cardiovascular diseases, pain, and many other conditions.
- **Scientific Relevance:** Studying how drugs act helps in understanding body functions such as nerve transmission, hormone regulation, and homeostasis.
- **Social and Ethical Perspective:** While therapeutic use can save lives, misuse or addiction can lead to severe health and social problems.

Thus, pharmacological drugs occupy a unique position in human biology: they can be powerful tools for healing when used wisely, but equally dangerous when abused or misunderstood.

SLO [B-12-T-01]

Explain the role of biological warfare with examples.

DRUG DISCOVERY AND DEVELOPMENT PROCESS

A **drug** is a chemical substance that alters normal physiological functions of the body. In medicine, drugs are developed to **diagnose, prevent, or treat diseases**. The discovery and development of new drugs is a **complex, multi-phase process** that can take a decade or more, requiring collaboration across biology, chemistry, pharmacology, and regulatory sciences. The ultimate goal is to deliver **safe, effective, and high-quality medications** for human use.

1. Drug Discovery

The **drug discovery stage** is the foundation of the entire development process. Its purpose is to *identify biologically active compounds* that can influence the course of a disease. This stage combines modern tools of **molecular biology, biochemistry, genetics, and computational sciences** to search for promising drug candidates. Since only a small fraction of potential compounds ever become medicines, drug discovery is **both challenging and highly selective**.



The main steps include:

- **Target Identification:** The process begins by identifying a **biological target**, such as a protein, enzyme, ion channel, receptor, or nucleic acid that plays a crucial role in the disease mechanism. For example, in bacterial infections, enzymes essential for cell wall synthesis may serve as drug targets. In cancer, proteins that control cell division may be selected. A good target is one that is **disease-specific** and **"druggable"**, meaning it can be modulated by a chemical compound.
- **Target Validation:** Once a potential target is identified, researchers must confirm that altering it will actually produce a therapeutic effect. This involves laboratory experiments, genetic studies, and use of animal models to show that blocking or activating the target improves the disease condition. For example, validation of HIV protease as a target paved the way for protease inhibitor drugs used in antiretroviral therapy.
- **Hit Identification:** In this step, scientists look for molecules that show initial activity against the target. This is done using **high-throughput screening**, a technique that tests thousands of compounds from large chemical libraries in a short time. Molecules that show promising interactions are called **"hits."** Increasingly, computer-based methods such as **virtual screening** and **molecular docking** are also used to predict how compounds will interact with the target, saving both time and resources.
- **Hit-to-Lead Optimization:** Initial hits are rarely suitable as drugs because they may lack potency, have unwanted side effects, or be unstable in the body. Therefore, chemists modify the chemical structure of hits to create more effective compounds known as **"leads."** This process focuses on **improving**:
 - **Potency:** ensuring the compound strongly affects the target at low doses.
 - **Specificity:** reducing interaction with unintended targets to minimize side effects.
 - **Pharmacokinetics:** optimizing how the drug is absorbed, distributed, metabolized, and excreted (ADME properties).
 - **Safety and Toxicity:** ensuring that the compound does not cause harmful effects in cells or tissues.

The outcome of this stage is a small number of **lead compounds** with the highest potential to progress into **preclinical development**. These leads are then studied in greater depth for their safety, stability, and suitability for human use.

2. Preclinical Development

Once **lead compounds** have been identified and optimized, they must be carefully studied before being tested in humans. This stage is called **preclinical development**. Its purpose is to evaluate the **safety, biological activity, and drug-like properties** of the compound under controlled laboratory conditions. Preclinical studies usually take several years and provide the scientific evidence required for regulatory authorities to permit human trials.

The main components are:

- **In Vitro Studies:** These are experiments performed **outside a living organism**, usually on cultured cells or isolated tissues. In vitro studies help scientists understand:
 - How the compound interacts with its biological target.
 - Its effectiveness in blocking or stimulating a disease pathway.
 - Potential toxic effects on human cells. For example, cancer drug candidates may be tested on tumor cell lines to measure whether they can inhibit abnormal cell growth.
- **In Vivo Studies:** These involve experiments in **living animal models** (such as mice, rats, or primates). In vivo studies provide information about:
 - **Pharmacokinetics (PK):** how the drug is absorbed, distributed, metabolized, and excreted in the body.
 - **Pharmacodynamics (PD):** the biological and physiological effects of the drug, including its mechanism of action.
 - **Safety and efficacy:** whether the compound improves disease symptoms in animals without causing unacceptable harm.
- **Toxicology Studies:** Every potential drug must be tested for **toxicity before human use**. Toxicology studies determine:
 - **Acute toxicity:** harmful effects after a single or short-term exposure.
 - **Chronic toxicity:** effects after repeated or long-term use.
 - **Safe dosage ranges:** the highest dose that can be given without significant adverse effects.
 - This ensures that the drug candidate has a favorable **therapeutic index** (the ratio between effective dose and toxic dose).

- **Formulation Development:** Even if a compound is effective, it must be delivered to the body in a practical and stable form. In this step, scientists design suitable **dosage forms** such as tablets, capsules, syrups, injections, or topical creams. They also consider:
 - **Stability** (shelf life and resistance to degradation).
 - **Bioavailability** (the fraction of the drug that reaches circulation).
 - **Delivery mechanisms** (e.g., oral vs. injectable).
 For instance, some drugs cannot survive the acidic environment of the stomach and must be given as coated tablets or injections.

Preclinical development ensures that only the most **promising and safe drug candidates** progress to human testing. Although many compounds fail at this stage, it serves as a critical filter to protect human volunteers in later clinical trials.

3. Clinical Development

If a drug candidate successfully passes **preclinical testing**, the next step is to evaluate it in humans. Before trials begin, an application such as the **Investigational New Drug (IND)** is submitted to regulatory authorities (e.g., FDA in the USA, DRAP in Pakistan) for approval. Only after this clearance can **clinical development**—human testing—proceed.

Clinical trials are conducted in **phases**, each designed to answer specific questions about the drug's safety, effectiveness, and appropriate use.

- **Phase I: Safety and Dosage:**
 - **Participants:** 20–100 healthy volunteers (sometimes patients, if the drug is very risky, e.g., for cancer).
 - **Purpose:** To assess **safety, tolerability, dosage range, and metabolism** of the drug in humans.
 - **Key Question:** *Is the drug safe enough to be given to people, and how does the human body process it?*
- **Phase II: Efficacy and Side Effects:**
 - **Participants:** 100–500 patients with the target disease.
 - **Purpose:** To test the therapeutic effectiveness of the drug, refine the correct dose, and monitor short-term side effects.
 - **Key Question:** *Does the drug work in patients, and what are its risks?*
- **Phase III: Large-Scale Testing:**
 - **Participants:** 1,000–3,000 patients across multiple hospitals and countries.
 - **Purpose:** To **confirm effectiveness**, compare with existing treatments, and identify **less common or long-term side effects**.
 - **Key Question:** *Is the new drug better and safer than current standard treatments?*
 - Successful Phase III results usually form the basis for requesting **regulatory approval**.
- **Phase IV: Post-Marketing Surveillance:**
 - Conducted after the drug is approved and available to the public.
 - **Purpose:** To track long-term safety and effectiveness in a much larger, diverse population. Rare adverse reactions, interactions with other medicines, or previously unknown risks are often discovered at this stage.
 - Sometimes Phase IV studies also test new uses or formulations of the drug.

Clinical development is the **most critical stage** of drug testing, as it provides the final evidence of whether a compound can become a safe and effective medicine. Although the process is long, expensive, and sometimes results in failure, it ensures that only the best candidates are introduced for patient use.

4. Regulatory Review

After a drug successfully completes **preclinical and clinical development**, it cannot be marketed immediately. Before reaching patients, it must pass through a stage called **regulatory review**, where independent authorities carefully examine all the collected data. This step ensures that only medicines that meet the highest standards of **safety, efficacy, and quality** are approved for public use.

The process involves two major steps:

- **New Drug Application (NDA):**
 - The pharmaceutical company compiles all data from **laboratory studies, animal testing, and clinical trials** into a comprehensive document known as a **New Drug Application (NDA)** (or an equivalent dossier in other countries).
 - The NDA includes details about:
 - The drug's **mechanism of action**.
 - Results of preclinical and clinical trials.



- Safety and toxicity profiles.
- Proposed dosage forms and manufacturing processes.
- Labeling information for doctors and patients.
- This submission is made to regulatory authorities such as the **FDA (USA), EMA (Europe), or DRAP (Pakistan)**.
- **Evaluation by Regulatory Authorities:**
 - Expert committees of scientists, physicians, and pharmacologists critically examine the application.
 - They review whether:
 - The drug demonstrates **real therapeutic benefit** compared to existing treatments.
 - The risks and side effects are **acceptable** in relation to its benefits.
 - The drug can be consistently manufactured to meet quality standards.
 - If the evidence is convincing, approval is granted, and the medicine can be marketed. If not, the application may be rejected or returned for further testing.

The **regulatory review** process acts as the final safeguard between experimental drugs and patient use. By ensuring rigorous evaluation, it protects public health and maintains trust in modern medicine.

5. Post-Marketing Surveillance

Approval by regulatory authorities does not mean that a drug is completely free of risk. Once a medicine enters the market, it is used by **millions of people** in diverse settings, including patients with multiple health conditions, children, the elderly, and pregnant women—groups that may not have been fully represented in clinical trials. For this reason, **post-marketing surveillance** (also called **Phase IV studies**) is essential to ensure continued safety and effectiveness.

The key elements include:

- **Monitoring:**
 - Long-term safety tracking is carried out through reporting systems where doctors, pharmacists, and patients can submit information about side effects.
 - This helps identify **rare, delayed, or unexpected adverse reactions** that may not have appeared in earlier trials because of limited sample sizes or shorter duration.
 - For example, some painkillers and anti-diabetic drugs have been withdrawn from the market after serious cardiovascular side effects were discovered post-approval.
- **Additional Studies:**
 - Post-marketing research may investigate:
 - **New therapeutic uses** of the drug (e.g., aspirin, originally developed as a painkiller, later found to reduce the risk of blood clots).
 - **New dosage forms** such as slow-release tablets, syrups, or injections.
 - **Combination therapies**, where the drug is tested with other medicines to improve effectiveness or reduce side effects.
 - These studies can expand the usefulness of a drug while ensuring patient safety.

Post-marketing surveillance is the **final layer of protection** in the drug development process. It ensures that medicines remain safe and effective under real-world conditions, and it provides valuable data that can guide improvements, new indications, or even withdrawal of unsafe products.

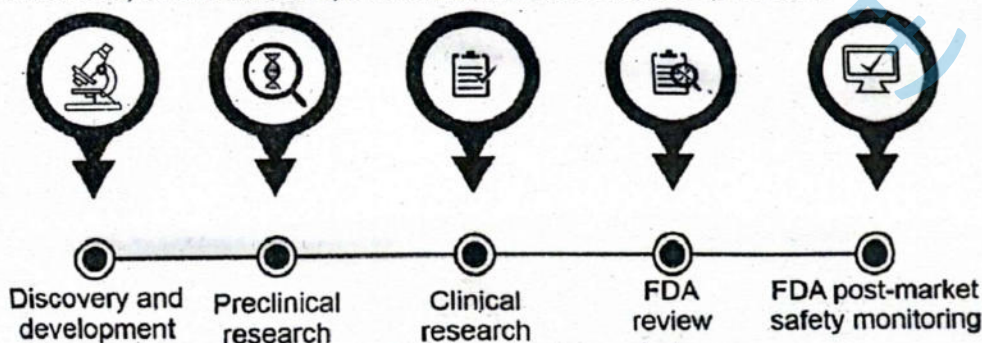


Figure: The five drug development phases

The drug discovery and development process is **lengthy, expensive, and highly selective**, with many candidates failing along the way. However, it ensures that only medicines meeting rigorous standards of **safety, efficacy, and quality** reach patients, protecting public health.

CLASSES OF ANTIBIOTICS AND THEIR MODE OF ACTION

Antibiotics are chemical substances, either naturally produced by microorganisms or synthetically manufactured, that inhibit the growth of or kill other microorganisms, especially bacteria. They are grouped into different classes based on their chemical structure and mechanism of action.

Here are four important classes:

1. Penicillins

- **Definition:** Penicillins were the **first true antibiotics** discovered, marking a revolution in medicine. They were originally derived from the mold *Penicillium notatum* by **Alexander Fleming in 1928** and later mass-produced during World War II. They belong to the **β -lactam class of antibiotics** (so named because they contain a **β -lactam ring** – a special four-sided chemical structure essential for their action).
- **Mode of Action:** Penicillins specifically target the **bacterial cell wall** (the tough outer layer that protects bacterial cells, not present in human cells, which makes the drug safe for us).
 - They block the enzyme **transpeptidase** (also called **penicillin-binding protein, PBP** – a bacterial enzyme that helps “stitch” the wall together).
 - Normally, this enzyme cross-links **peptidoglycan chains** (peptidoglycan = a mesh-like sugar + protein material that gives bacteria their rigid shape).
 - Without cross-linking, the cell wall becomes **weak and leaky**.
 - Under pressure, the bacteria **burst and die**.
 - This killing effect makes penicillins **bactericidal** (they kill bacteria outright, instead of just slowing them down).
 - They are especially effective against **Gram-positive bacteria** (bacteria with a thick cell wall, e.g., *Streptococcus*).
- **Examples and Uses:**
 - **Penicillin G:** The original penicillin; effective against pneumonia, strep throat, and syphilis.
 - **Amoxicillin:** A semi-synthetic version (chemically improved form), stable and effective for ear, throat, and urinary tract infections.
 - **Ampicillin:** Broader spectrum than Penicillin G, can act against some Gram-negative bacteria (Gram-negative = bacteria with thinner walls but an extra outer membrane, e.g., *E. coli*).
- **Clinical Importance:**
 - Before penicillin, simple infections like pneumonia or infected wounds often caused death.
 - The discovery of penicillin turned once-deadly diseases into treatable conditions and laid the foundation for modern antibiotics.

2. Tetracyclines

- **Definition:** Tetracyclines are a class of **broad-spectrum antibiotics** (meaning they can act against a wide range of bacteria, both **Gram-positive** and **Gram-negative**).
 - They were first discovered in the 1940s and remain widely used because of their effectiveness.
 - They are particularly useful against **respiratory infections, acne, urinary tract infections, and some diseases caused by unusual bacteria** (e.g., *Chlamydia*, *Rickettsia*, and *Mycoplasma*).
- **Mode of Action:** Unlike penicillins (which attack the bacterial wall), tetracyclines target **bacterial protein production**, which is essential for survival.
 - They bind to the **30S ribosomal subunit** (a smaller part of the bacterial ribosome – the ribosome is like a “protein-making factory” inside the cell).
 - This blocks the attachment of **aminoacyl-tRNA** (the “carrier molecules” that bring amino acids, the building blocks of proteins, to the ribosome).
 - Without this step, the **elongation of the protein chain** (the process of making longer proteins) *is stopped*.
 - Since bacteria cannot make proteins, they **stop growing and multiplying**.
 - This makes tetracyclines **bacteriostatic** (they stop bacterial growth rather than killing bacteria directly).

- **Examples and Uses:**
 - **Tetracycline:** The original drug, used for respiratory infections and acne.
 - **Doxycycline:** A newer, more stable version; widely used for malaria prevention, typhus, Lyme disease, and sexually transmitted infections (STIs).
 - **Minocycline:** Used for acne and some resistant infections.
- **Clinical Importance:**
 - Tetracyclines are valued for treating infections where other antibiotics may not work, especially **atypical bacteria** (those that don't have the usual bacterial cell wall or structure).
 - They are also used in combination with other antibiotics in complex infections.

3. Fluoroquinolones

- **Definition:** Fluoroquinolones are a class of **fully synthetic antibiotics** (unlike penicillins or tetracyclines, which came from natural sources, these are man-made in laboratories).
 - They were developed in the 1980s to provide **broad-spectrum coverage**, meaning they can fight a wide variety of bacteria, including both **Gram-positive** and **Gram-negative** species.
 - They are especially effective against bacteria that cause **respiratory infections, urinary tract infections, gastrointestinal infections, and some hospital-acquired infections**.
- **Mode of Action:** Fluoroquinolones work by targeting the **bacterial DNA machinery**—the system bacteria use to copy and repair their genetic material.
 - They block two key enzymes:
 - **DNA gyrase** (helps unwind the tightly coiled DNA so it can be copied).
 - **Topoisomerase IV** (helps separate newly made DNA strands during cell division).
 - Without these enzymes, bacteria cannot replicate or repair their DNA.
 - This causes DNA damage to build up, leading to bacterial cell death.
 - Because they directly kill bacteria rather than just stopping growth, fluoroquinolones are bactericidal.
- **Examples and Uses:**
 - **Ciprofloxacin:** Effective against urinary tract infections, gastrointestinal infections (like typhoid), and anthrax.
 - **Levofloxacin:** Often used for pneumonia, sinus infections, and bronchitis.
 - **Moxifloxacin:** Stronger activity against Gram-positive bacteria; used for respiratory infections and some eye infections.
- **Clinical Importance:**
 - Fluoroquinolones are popular because they are **well absorbed orally** (patients can take tablets instead of injections).
 - They also **penetrate deep into tissues**, making them useful for bone, joint, and respiratory infections.
 - They are considered "**reserve antibiotics**" in some cases, used when other drugs fail.
- **Limitations and Side Effects:**
 - **Resistance:** Overuse has led to resistant strains of bacteria (especially *E. coli* and *Pseudomonas*).
 - **Tendon damage:** Rarely, they can cause **tendonitis or tendon rupture** (tearing of connective tissues like the Achilles tendon).
 - **Nervous system effects:** Can cause headaches, dizziness, or sleep problems.
 - **Heart rhythm changes:** Some fluoroquinolones can disturb electrical activity in the heart (QT prolongation).
 - Because of these risks, they are often reserved for **serious infections**.

4. Sulfonamides

- **Definition:** Sulfonamides, also called **sulfa drugs**, were the **first widely used synthetic antibiotics** (made entirely in the laboratory rather than derived from natural organisms).
 - They were introduced in the 1930s, before penicillin became available, and saved countless lives from bacterial infections.
 - Even today, they are still used—often in **combination with other antibiotics**—to treat a variety of infections.
- **Mode of Action:** Sulfonamides interfere with how bacteria make **folic acid** (vitamin B9), which is essential for their survival.
 - They act as **competitive inhibitors** (meaning they "compete" with the natural substance for the enzyme's active site, like putting the wrong key in a lock).
 - The enzyme they block is **dihydropteroate synthase** (an enzyme bacteria use to start building folic acid).
 - Without folic acid, bacteria cannot produce **DNA** (the genetic material) or **proteins**, both of which are essential for growth and division.



- Humans are unaffected because we do not make folic acid ourselves—we get it from food—so sulfonamides specifically target bacteria.
- The effect is **bacteriostatic** (they don't directly kill bacteria, but stop them from multiplying, giving the immune system time to eliminate them).
- **Examples and Uses:**
 - **Sulfamethoxazole:** Commonly used in combination with another drug, Trimethoprim (the combination is called **Co-trimoxazole** or **Septtrin**).
 - This combination blocks two different steps in folic acid production, making the treatment much more effective.
 - They are used for **urinary tract infections, respiratory infections, intestinal infections, and some opportunistic infections in immunocompromised patients (like those with HIV/AIDS).**
- **Clinical Importance:**
 - Sulfonamides were the first “miracle drugs” before penicillin and played a key role in reducing deaths from infections in the pre-antibiotic era.
 - Their discovery marked the beginning of the modern era of antimicrobial therapy.

Summary Table

Class	Target/Action	Effect on Bacteria	Example
Penicillins	Inhibit cell wall synthesis (block transpeptidase)	Bactericidal	Penicillin G, Amoxicillin
Tetracyclines	Bind 30S ribosome → block protein synthesis	Bacteriostatic	Doxycycline
Fluoroquinolones	Inhibit DNA gyrase & topoisomerase IV → block DNA replication	Bactericidal	Ciprofloxacin
Sulfonamides	Inhibit folic acid synthesis (enzyme: dihydropteroate synthase)		

SLO [B-12-T-04]

Define antivirals and antiretrovirals.

ANTIVIRALS AND ANTIRETROVIRALS

Antiviral drugs are designed to treat infections caused by viruses. Unlike bacteria, viruses rely entirely on the host's cellular machinery for replication, making them more difficult to target.

Antivirals

Antiviral drugs inhibit viral replication by interfering with various stages of the viral life cycle, such as attachment, entry, genome replication, or protein synthesis. These drugs are used against a broad range of viruses including influenza viruses, herpesviruses, and hepatitis viruses.

Antiretrovirals

Antiretrovirals are a specialized class of antiviral drugs specifically used to combat **retroviruses**, particularly **HIV (Human Immunodeficiency Virus)**. Retroviruses use RNA as their genetic material and rely on the enzyme reverse transcriptase for replication.

Different classes of antiretroviral drugs act at various stages of the viral life cycle, including:

- Entry inhibitors
- Reverse transcriptase inhibitors
- Integrase inhibitors
- Protease inhibitors

These drugs are often used in combination therapy (HAART – Highly Active Antiretroviral Therapy) to prevent resistance and manage viral load effectively.

DO YOU KNOW?

All antiretrovirals are antivirals but not all antivirals are antiretrovirals. Retroviruses, helping to control HIV infection and reduce the risk of progression to AIDS.



ADVANTAGES OF MONOCLONAL ANTIBODIES

Monoclonal antibodies are laboratory-produced molecules that mimic the immune system's ability to recognize and neutralize specific pathogens or abnormal cells. These antibodies are derived from a single clone of cells and bind to a single specific antigen.

Mechanism of Action

Monoclonal antibodies are engineered to recognize and bind very precisely to a **specific antigen** (a unique marker, usually a protein) found on the surface of **pathogens** (like viruses and bacteria) or **abnormal human cells** (like cancer cells). Once bound, they can act in multiple ways:

1. Direct Neutralization

- The antibody binds to a harmful molecule or pathogen and **blocks its activity**.
- Example: mAbs used in viral infections can stop viruses from attaching to host cells.

2. Immune System Activation

- By attaching to the target cell, mAbs act like a **"flag" or signal** to the immune system.
- This attracts immune cells (like macrophages and natural killer cells) to destroy the marked cell.

3. Complement Activation

- Binding of mAbs can trigger the **complement system** (a group of blood proteins that punch holes in cell membranes).
- This leads to the destruction of the targeted cell.

4. Blocking Growth Signals

- Some cancer cells grow uncontrollably because of signals from receptors on their surface.
- mAbs can **block these receptors**, preventing cancer cells from receiving "grow and divide" messages.

5. Delivery of Toxic Agents

- mAbs can be linked to toxins, chemotherapy drugs, or radioactive substances.
- Once the antibody finds its target, it delivers the toxic "payload" directly to diseased cells, sparing most healthy ones.

The mechanism of monoclonal antibodies is not limited to one function—they can **neutralize, tag, block, or directly kill diseased cells**. Their high precision makes them powerful tools against cancer, autoimmune diseases, and infections.

Advantages of Using Monoclonal Antibodies

1. High Specificity

- Monoclonal antibodies are designed to bind to **one specific antigen** only. They act like a "sniper bullet" instead of a "shotgun blast," hitting only the intended target.
- This precision greatly **reduces off-target effects** (unwanted interactions with healthy tissues).

2. Consistent Quality

- Because they are made in controlled laboratory systems, every batch of mAbs is nearly **identical in structure and function**. Like producing identical keys from the same mold—every copy works the same way.
- This ensures **reliable results** in both treatment and research.

3. Renewable Production

- Once scientists create a **hybridoma** (a fused cell that continuously makes antibodies) or a **recombinant cell line** (genetically engineered cell), it can produce unlimited amounts of the same antibody.

4. Versatile Applications

- **Diagnostics:** Used in pregnancy tests, HIV tests, COVID-19 rapid kits, and other diagnostic assays.

DO YOU KNOW?

The difference between monoclonal antibodies and polyclonal antibodies is in the names. "Mono" refers to one and "poly" refers to many. Monoclonal antibodies are clones of just one antibody, and they bind to one antigen only.

- **Therapy:** Effective against cancers (e.g., breast cancer, leukemia), autoimmune diseases (e.g., rheumatoid arthritis), and infectious diseases.
- **Research:** Widely used in molecular biology, drug development, and immunology experiments.

5. Engineering Flexibility

- Scientists can **genetically modify** mAbs to:
 - Increase their binding strength.
 - Reduce side effects.
 - Extend their half-life (how long they stay active in the body).

6. Humanization

- Originally, mAbs were made in mice, which sometimes triggered immune reactions in humans.
- Modern biotechnology allows scientists to **engineer mouse antibodies to closely resemble human antibodies**, reducing the risk of immune rejection.

7. Large-Scale Production

- Advances in **bioreactors and biotechnology** now allow mass production of monoclonal antibodies with high purity and safety standards.
- This makes them widely available for clinical use worldwide.

Monoclonal antibodies are superior to many traditional drugs because they are **highly specific, reliable, renewable, versatile, and customizable**. Their ability to be engineered and mass-produced has made them central tools in modern medicine, diagnostics, and research.

Monoclonal Antibodies vs Other Therapeutic Agents

1. Compared to Antibiotics

- **Antibiotics** act broadly on bacterial processes (e.g., cell wall synthesis, protein synthesis) and can affect a wide range of bacterial species.
- **Monoclonal antibodies (mAbs)**, however, are **precisely targeted** against one specific antigen, which reduces **off-target effects** (unwanted harm to beneficial bacteria or healthy cells).
- mAbs are also **less prone to resistance development**, because they target unique proteins rather than general bacterial machinery.
- **Layman's view:** Antibiotics are like "pesticides" that kill many bugs (good and bad), while mAbs are like "snipers" that only hit the exact harmful target.

2. Compared to Polyclonal Antibodies

- **Polyclonal antibodies** are natural mixtures produced by the immune system, binding to **multiple epitopes** (different parts of the same antigen).
- This leads to **variability** between batches and less precise targeting.
- **Monoclonal antibodies** bind to **only one epitope** (a single, unique spot on the antigen), ensuring **superior specificity and consistency**.
- **Layman's view:** Polyclonal antibodies are like a group of archers shooting arrows in different directions, while monoclonal antibodies are one expert archer hitting the bull's-eye every time.

3. Compared to Chemotherapy and Radiotherapy

- **Chemotherapy and radiotherapy** kill rapidly dividing cells, but they **cannot distinguish cancer cells from healthy dividing cells**, leading to side effects such as hair loss, nausea, and immune suppression.
- **Monoclonal antibody therapies** can selectively bind only to **cancer-specific antigens**, which allows:
 - Lower required dosages.
 - Reduced collateral damage to healthy tissues.
- **Layman's view:** Traditional therapies are like "carpet bombing" (hitting everything in the area), while mAbs are like "guided missiles" targeting only enemy bases.

4. Compared to Small-Molecule Drugs

- Many small-molecule drugs are rapidly cleared from the body and have **unpredictable pharmacokinetics** (how the body absorbs, distributes, metabolizes, and eliminates them).
- **Monoclonal antibodies, especially IgG type, have long half-lives** (~21 days), meaning they stay active in the body **much longer** and require less frequent dosing.
- Their action is also more predictable.

SCIENCE TITBITS

Monoclonal antibodies have been used in treatment of Cancer, Organ transplant rejection, Inflammatory and autoimmune disorders, including allergies, Infections, including COVID-19, Osteoporosis, Eye disorders, Migraines, High cholesterol, Nervous system disorders. In most cases, monoclonal antibodies are given mostly as intravenous (IV) solution injected right into your vein (sometimes referred to as an infusion).



1. **What is the first step in the drug discovery process?**
A. Hit Identification
B. Target Identification
C. Formulation Development
D. Phase I Trials
2. **What is the primary goal of hit-to-lead optimization?**
A. Reduce side effects
B. Improve cost-efficiency
C. Optimize drug properties
D. Test on humans
3. **Which study type uses cultured cells to assess drug efficacy?**
A. In vivo
B. In vitro
C. Toxicology
D. Clinical
4. **What is the purpose of in vivo studies in drug development?**
A. Assess effects in a petri dish
B. Test for regulatory approval
C. Observe drug effects in living organisms
D. Identify new diseases
5. **Which trial phase tests safety in 20–100 healthy volunteers?**
A. Phase I
B. Phase II
C. Phase III
D. Phase IV
6. **What does the New Drug Application (NDA) include?**
A. Financial projections
B. Only clinical trial results
C. All preclinical and clinical data
D. Animal study summaries
7. **Why are Phase IV clinical trials important?**
A. Confirm dosage
B. Reduce cost
C. Monitor long-term safety and rare side effects
D. Develop marketing strategies
8. **Which class of antibiotics inhibits cell wall synthesis?**
A. Fluoroquinolones
B. Sulfonamides
C. Penicillins
D. Tetracyclines
9. **How do tetracyclines inhibit bacterial growth?**
A. Disrupting DNA synthesis
B. Destroying membranes
C. Binding to 30S ribosomal subunit
D. Breaking lipid chains
10. **Which enzyme is targeted by fluoroquinolones?**
A. Reverse transcriptase
B. Protease
C. Topoisomerase IV
D. Ribosome synthase
11. **Sulfonamides disrupt which bacterial process?**
A. Cell wall formation
B. Folate synthesis
C. Protein folding
D. Lipid metabolism
12. **Why don't antibiotics work on viruses?**
A. Viruses have thick membranes
B. Viruses aren't alive
C. Viruses lack independent replication machinery
D. Viruses have resistant DNA
13. **What distinguishes retroviruses from others?**
A. Lack of nucleus
B. Use of RNA and reverse transcriptase
C. Infection of bacteria
D. Multicellular structure
14. **Which enzyme do reverse transcriptase inhibitors block?**
A. DNA polymerase
B. Protease
C. Reverse transcriptase
D. Helicase
15. **What is the combination antiretroviral strategy called?**
A. Monotherapy
B. HAART
C. Passive immunization
D. Retro-cycle
16. **Monoclonal antibodies are derived from:**
A. Blood plasma
B. Many B cells
C. A single B cell clone
D. T-cell clusters
17. **What enhances mAbs compatibility in humans?**
A. Crystallization
B. Polyclonality
C. Humanization
D. Protein folding
18. **Which field doesn't commonly use monoclonal antibodies?**
A. Diagnostics
B. Food preservation
C. Therapy
D. Research
19. **How do mAbs help treat cancer?**
A. Inhibit all dividing cells
B. Strengthen healthy tissue
C. Bind to cancer-specific antigens
D. Increase antibody count
20. **Which property makes mAbs more predictable than small molecules?**
A. Shorter half-life
B. Resistance development
C. Predictable pharmacokinetics
D. Natural sourcing

Answer Key

1. B	2. C	3. B	4. C	5. A	6. C	7. C	8. C	9. C	10. C
11. B	12. C	13. B	14. C	15. B	16. C	17. C	18. B	19. C	20. C



Q.1 Define the term 'drug' in pharmacology and state its uses.

Ans. In pharmacology, a **drug** is defined as a chemical substance, either natural or synthetic, that interacts with a biological system to produce a functional change. Drugs may act by enhancing, inhibiting, or modifying normal body functions or by interfering with disease-causing agents. Importantly, drugs are not only used for curing illnesses but also play roles in **diagnosis, treatment, prevention, and relief of symptoms**. For example, antibiotics cure infections, vaccines prevent diseases, and analgesics relieve pain. Drugs can also be used in diagnostic procedures such as imaging. Thus, a drug is a **therapeutic or diagnostic agent** designed to restore or maintain health.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; uses – 1 mark

Q.2 Explain the importance of target validation in drug discovery.

Ans. **Target validation** is a crucial step in the drug discovery process. It confirms that a chosen biological molecule, such as a receptor, enzyme, or gene, is directly involved in the disease process. First, target validation ensures that manipulating this target will yield the **desired therapeutic effect**, thereby reducing the chances of drug failure. Second, it helps prioritize promising drug targets so that **time, cost, and research efforts** are not wasted on ineffective ones. Third, validated targets provide a strong foundation for designing and testing drug candidates with higher chances of success. In this way, target validation improves the **efficiency, accuracy, and safety** of drug discovery.

MARKING SCHEME (3 Marks)

- Three significances – 1 mark each

Q.3 Differentiate between in vitro and in vivo studies in drug testing.

Ans. **In vitro studies** are experiments performed **outside a living organism**, typically in controlled laboratory environments such as petri dishes, test tubes, or cell cultures. They allow researchers to analyze drug effects on isolated cells, tissues, or biochemical systems, offering high control and reproducibility. However, they do not fully mimic the complexity of a whole organism. In contrast, **in vivo studies** are carried out **inside living organisms**, such as animals or humans. These studies test how a drug behaves within the complete biological system, including metabolism, absorption, and overall safety. Thus, in vitro studies provide **preliminary insights**, while in vivo studies reveal **real-world biological responses**.

MARKING SCHEME (3 Marks)

- Definition of in vitro – 1.5 marks; definition of in vivo – 1.5 marks

Q.4 Describe the function and mode of action of sulfonamide antibiotics.

Ans. Sulfonamide antibiotics were the first widely used antimicrobial drugs and remain important in medicine. Their primary function is to **inhibit bacterial growth** by interfering with DNA synthesis. They achieve this by competitively inhibiting the enzyme responsible for converting **para-aminobenzoic acid (PABA)** into **folic acid**, a molecule essential for nucleic acid production in bacteria. Since humans obtain folic acid from diet and do not synthesize it, sulfonamides act with **selective toxicity**, harming bacteria without damaging human cells. By blocking folic acid synthesis, these antibiotics stop bacterial replication and spread, helping the immune system eliminate infection. Hence, sulfonamides are **bacteriostatic agents** effective against a wide range of bacteria.

MARKING SCHEME (3 Marks)

- Mode of action – 3 marks

Q.5 State the purpose of Phase III clinical trials in drug development.

Ans. **Phase III clinical trials** are the final and largest stage of human drug testing before regulatory approval. Their primary purpose is to **evaluate the overall effectiveness** of the drug in treating the target disease within large and diverse patient populations. They also **monitor side effects and safety** by detecting both common and rare adverse reactions that may not appear in earlier trials. In addition, Phase III trials **compare the new drug with existing treatments or placebo**, helping determine whether it offers superior benefits. Results from Phase III provide critical evidence for final approval by drug regulatory authorities, ensuring that only safe and effective drugs reach the market.

MARKING SCHEME (3 Marks)

- Effectiveness – 1 mark; side effect monitoring – 1 mark; comparison with treatments – 1 mark



Q.6 Define the term 'hit identification' in drug discovery and state its advantage.

Ans. In drug discovery, **hit identification** refers to the process of finding chemical compounds, known as "hits," that interact with a chosen biological target and show promising therapeutic potential. This is achieved by screening large libraries of natural products, synthetic molecules, or computationally designed compounds. The advantage of hit identification is that it provides a **starting point for drug development**, allowing researchers to focus only on those molecules that demonstrate measurable biological activity. Without this stage, drug discovery would be inefficient, as countless inactive molecules would waste resources. Thus, hit identification serves as the **gateway to lead optimization and preclinical research**.

MARKING SCHEME (3 Marks)

- Definition – 2 marks; advantage – 1 mark

Q.7 Explain the importance of Phase IV clinical trials in drug development.

Ans. **Phase IV trials**, also called **post-marketing surveillance**, are conducted after a drug has been approved and released to the public. Their importance lies in detecting **rare or long-term side effects** that may not be evident during earlier, smaller-scale trials. They also assess the drug's **real-world effectiveness** across diverse patient groups, including those with multiple diseases or on other medications. Furthermore, Phase IV trials provide information about **drug interactions, long-term safety, and cost-effectiveness**. This ensures that the drug continues to be safe and beneficial for widespread use. In some cases, Phase IV findings may even lead to withdrawal of unsafe drugs or adjustment of usage guidelines.

MARKING SCHEME (3 Marks)

- Purpose of Phase IV trials – 3 marks

Q.8 Describe step by step the mode of action of fluoroquinolones.

Ans. Fluoroquinolones are a class of broad-spectrum antibiotics that act by **disrupting bacterial DNA replication**. Their mode of action can be explained step by step:

MARKING SCHEME (3 Marks)

- Step-by-step mode of action – 3 marks

1. They bind to and inhibit the bacterial enzyme **DNA gyrase**, which normally introduces negative supercoils into DNA.
2. They also inhibit **topoisomerase IV**, an enzyme required to separate replicated DNA strands during cell division.
3. Without these enzymes, bacterial DNA replication and repair are blocked, leading to **fragmentation of DNA**.
4. As a result, bacteria cannot divide or survive, leading to **cell death**. Since these enzymes are unique to bacteria, fluoroquinolones act selectively without harming human cells.

Q.9 Explain how sulfonamides work selectively in bacterial cells.

Ans. **Sulfonamides** are bacteriostatic antibiotics that act selectively on bacterial cells by interfering with folic acid metabolism. Their mode of action can be described in steps:

MARKING SCHEME (3 Marks)

- Step-by-step mode of action – 3 marks

1. Bacteria synthesize folic acid from **para-aminobenzoic acid (PABA)** using the enzyme **dihydropteroate synthase**.
2. Sulfonamides, being structural analogues of PABA, competitively inhibit this enzyme, preventing folic acid formation.
3. Without folic acid, bacteria cannot synthesize nucleic acids and proteins, leading to growth inhibition.
4. This selectivity arises because **human cells do not synthesize folic acid**; instead, they obtain it from diet. Therefore, sulfonamides specifically target bacterial metabolism without harming human tissues.

Q.10 State the therapeutic advantages of monoclonal antibodies (mAbs) over traditional drugs.

Ans. Monoclonal antibodies (mAbs) are laboratory-produced molecules that can precisely bind to specific antigens. Their therapeutic advantages over traditional small-molecule drugs are significant. First, they have **high specificity**, targeting only diseased cells or molecules, which minimizes damage to healthy tissues. Second, mAbs generally cause **fewer side effects**, since they do not interfere with **unrelated biological pathways**. Third, they have a **longer circulation time in the bloodstream**, requiring less frequent dosing and improving patient compliance. Additionally, they can be engineered to **modulate the immune system** or block specific signaling pathways, offering advanced treatment options for cancer, autoimmune disorders, and infections.

MARKING SCHEME (3 Marks)

- Three main advantages – 1 mark each



Exercise

I

Multiple Choice Questions (MCQs)

❖ Select the correct Answer.

1. What is the primary goal of target identification in drug discovery?
A. To determine the chemical composition of a drug.
B. To identify the biological origin of a disease and potential targets for intervention.
C. To evaluate the safety and efficacy of a drug in animal models.
D. To optimize the synthetic route for a drug's production.
2. Which clinical trial phase involves testing a drug on a small group of healthy volunteers to assess its safety and determine the appropriate dosage?
A. Phase I
B. Phase II
C. Phase III
D. Phase IV
3. The pharmacokinetic properties of a drug are mainly governed by the physicochemical properties of the drug such as it's:
A. Solubility in aqueous solutions
B. Size and structure
C. Ability to form hydrogen bonds
D. All of these
4. What is the main purpose of phase IV testing?
A. Determine therapeutic window
B. Determine unreported toxicity
C. Determine variation in absorption
D. Determine tolerability
5. Which among the following antibiotics function as protein synthesis inhibitors?
A. penicillin
B. tetracycline
C. cefotaxime
D. trimethoprim
6. Which of the following techniques is best for measuring the effectiveness of antibiotics on bacteria?
A. Physical testing
B. Chemical testing
C. Biological testing
D. Combined physical and chemical testing
7. Which point in the replication cycle appears most easily blocked by antivirals?
A. Virus absorption
B. Virus penetration
C. Virus RNA and DNA replication
D. Exit of viruses from the cell
8. Which is one of the main current goals of antiretroviral treatment?
A. to suppress the viral load
B. to cure HIV infection
C. to allow the patient to have a pain-free death
D. to prevent repeated infection with HIV
9. Which of the following is easily blocked by antivirals?
A. Virus penetration
B. Nucleic acid replication
C. Virus absorption
D. Removal of the virus from the cell
10. Which of the following is the primary reason why antiretroviral therapy (ART) cannot cure HIV?
A. ART is not effective in reducing the viral load.
B. HIV can hide within immune cells, forming latent reservoirs.
C. ART drugs are toxic and cause severe side effects.
D. HIV mutates too rapidly to be targeted by ART.
11. Which of the following cannot be treated by antiviral drugs?
A. tuberculosis
B. smallpox
C. hepatitis
D. warts
12. What is the main advantage of monoclonal antibodies over polyclonal antibodies?
A. greater specificity and constancy
B. lower cost production
C. longer shelf life
D. ability to target multiple antigens
13. How monoclonal antibodies are administered to patients?
A. topically
B. intravenously
C. inhalation
D. Orally
14. What is the main difference between Phase III and Phase IV clinical trials?
A. Phase III tests the drug's efficacy, while Phase IV focuses on long-term safety.
B. Phase III studies involve a small number of patients, while Phase IV studies involve a large number of patients.
C. Phase III trials are conducted in animals, while Phase IV trials are conducted in humans.
D. Phase III trials are conducted in one country, while Phase IV trials are conducted in multiple countries.



15. What is the purpose of a New Drug Application?
 A. To obtain permission to conduct clinical trials on a new drug.
 B. To request a patent for a new drug.
 C. To seek regulatory approval to market a new drug.
 D. To conduct post-marketing surveillance on a new drug.
16. Antibodies are:
 A. prostaglandins B. steroids
 C. lipoproteins D. glycoproteins
17. What are monoclonal antibodies?
 A. Antibodies produced by a single type of immune cell.
 B. Antibodies derived from multiple individuals.
 C. Antibodies with multiple antigen-binding sites.
 D. Antibodies found in the blood of newborn
18. What is the main advantage of monoclonal antibodies over polyclonal antibodies?
 A. Greater specificity and consistency
 B. Low cost production
 C. Ability to target multiple antigens simultaneously
 D. Longer shelf life.
19. The science which deals with the drug and their action on human body is called:
 A. Physiology B. Pathology
 C. Pharmacology D. Microbiology
20. The study of absorption, distribution, metabolism and excretion of drug is known as:
 A. Pharmacy B. Pharmacokinetics
 C. Pharmacodynamics D. Pharmacopoeia

Answer Key

1. B	2. A	3. D	4. B	5. B	6. C	7. C	8. A	9. B	10. B
11. A	12. A	13. B	14. A	15. C	16. D	17. A	18. A	19. C	20. B

II

Short Answer Questions

Q.1 Define the following:

(a) Pharmacology

Ans. Pharmacology is the branch of science that studies how drugs and chemical substances interact with living organisms. It involves understanding the **mechanisms of drug action, therapeutic effects, side effects, and toxicity**, as well as the absorption, distribution, metabolism, and excretion of drugs. Pharmacology forms the foundation for developing new medicines and for ensuring their safe and effective use in treating diseases.

MARKING SCHEME (3 Marks)

- Definition – 2 marks, significance – 1 mark

(b) Drug

Ans. A drug is any **natural or synthetic chemical substance** that alters the physiological or biochemical processes in a living organism. Drugs are used for **diagnosis, treatment, prevention, or cure of diseases**, and they act by interacting with specific molecular targets such as enzymes, receptors, or ion channels. Drugs may produce **therapeutic or toxic effects** depending on the dose and context.

MARKING SCHEME (3 Marks)

- Definition – 2 marks, example of use – 1 mark

(c) Drug Therapy

Ans. Drug therapy, or pharmacotherapy, refers to the use of drugs to **manage, alleviate, or cure diseases**. It is a central component of **modern medicine, relying on pharmacological knowledge** to select the appropriate drug, dose, and duration for each patient. Drug therapy **aims to restore health, reduce symptoms, prevent complications, and improve quality of life**.

MARKING SCHEME (3 Marks)

- Definition – 2 marks, purpose – 1 mark

(d) Antibiotics

Ans. Antibiotics are chemical agents that either **kill bacteria (bactericidal)** or **inhibit their growth (bacteriostatic)**. They are specifically used to treat **bacterial infections** by targeting structures or processes unique to bacteria, such as **cell wall synthesis, protein synthesis, or DNA replication**. Antibiotics do not affect viruses and must be used judiciously to prevent resistance.

MARKING SCHEME (3 Marks)

- Definition – 2 marks, mechanism / target – 1 mark



(e) Antivirals

Ans. Antivirals are drugs designed to **prevent the replication of viruses within host cells**. They do not directly kill viruses but **interfere with the viral life cycle**, such as inhibiting attachment, uncoating, genome replication, protein synthesis, or viral release. Antivirals reduce **severity, duration, and transmission** of viral infections and are used against diseases like influenza, herpes, and hepatitis.

MARKING SCHEME (3 Marks)
• Definition – 2 marks, mechanism – 1 mark

(f) Antiretrovirals

Ans. Antiretrovirals are a specialized class of antivirals used to **treat infections caused by retroviruses**, such as HIV. They inhibit viral replication by targeting **reverse transcriptase, integrase, protease, or viral entry**, thereby reducing viral load, preserving immune function, and delaying disease progression. While they **do not cure HIV**, they significantly improve patient outcomes.

MARKING SCHEME (3 Marks)
• Definition – 2 marks, mechanism – 1 mark

(g) Monoclonal Antibodies

Ans. Monoclonal antibodies are **laboratory-engineered molecules** that bind to **specific antigens**. They are highly selective and are used in **treating cancers, autoimmune disorders, and infectious diseases**. Their high specificity minimizes side effects, and they can be designed to target **specific immune or signaling pathways**, making them powerful tools in modern therapeutics.

MARKING SCHEME (3 Marks)
• Definition – 2 marks, therapeutic use – 1 mark

Q.2 What is a 'pharmacological drug'?

Ans. A pharmacological drug is a chemical substance of known structure that produces a measurable biological effect when administered to a living organism. These effects may be therapeutic, preventive, or diagnostic. Drugs can be derived from **natural sources** such as plants and microorganisms or **synthesized chemically** in laboratories. They interact with **specific molecular targets** such as receptors, enzymes, or ion channels, modifying normal body functions or disease processes. Pharmacological drugs are essential in **diagnosing diseases** (e.g., contrast agents in imaging), **treating infections** (e.g., antibiotics), **curing chronic conditions** (e.g., insulin for diabetes), or **preventing illness** (e.g., vaccines). Thus, they form the backbone of **modern medicine and healthcare**.

MARKING SCHEME (3 Marks)
• Definition – 2 marks; uses – 1 mark

Q.3 Write the four major steps of drug discovery.

Ans. Drug discovery is a structured process that involves several stages:

- 1. Target identification and validation** – Scientists identify a biological molecule (such as a protein or receptor) involved in a disease and confirm that altering its activity may have a therapeutic effect.
- 2. Lead compound identification** – A compound is discovered, often through screening libraries of chemicals or natural products, that shows activity against the chosen target.
- 3. Lead optimization** – The identified molecule is chemically modified to improve its potency, stability, safety, and ability to reach the target effectively.
- 4. Preclinical testing** – Laboratory and animal studies evaluate the compound's safety, toxicity, and pharmacological properties before advancing to human trials.

MARKING SCHEME (4 Marks)
• 1 mark each for four steps – 4 marks

Q.4 What are the steps of preclinical development of drugs?

Ans. **Preclinical development** is a critical stage that occurs before human testing begins. It focuses on evaluating whether a potential drug is safe and effective enough for clinical use. Laboratory experiments and **animal studies** are conducted to assess:

MARKING SCHEME (3 Marks)
• Explanation of preclinical testing steps – 3 marks

- **Safety and toxicity** – identifying harmful effects and safe dosage ranges.
- **Pharmacokinetics** – studying how the body absorbs, distributes, metabolizes, and excretes the drug.
- **Pharmacodynamics** – investigating how the drug interacts with its target and what biological effects it produces.
- These studies also explore **organ-specific toxicity** and long-term safety. The overall goal is to generate reliable data to support an application for clinical trials in humans.



Q.5 Describe clinical development of drugs.

Ans. Clinical development involves testing a drug candidate in human volunteers and patients through a sequence of phases to ensure safety and effectiveness.

- Phase I trials test the drug on a small group of healthy volunteers to determine safe dosage levels and identify side effects.
- Phase II involves patients with the target disease, focusing on the drug's effectiveness and refining the optimal dose.
- Phase III trials include large populations across multiple centers, comparing the new drug with existing treatments while monitoring long-term safety and efficacy.
- After successful results, data is submitted to regulatory authorities for approval. Only after this stage can the drug enter the market for medical use.

MARKING SCHEME (3 Marks)

- Description of phased testing – 3 marks

Q.6 Name the five drug development phases.

Ans. The five phases of drug development are:

1. Drug Discovery – identification of active compounds and targets.
2. Preclinical Development – laboratory and animal testing for safety and activity.
3. Clinical Development – human trials in multiple phases (I–III).
4. Regulatory Review – evaluation by authorities for approval.
5. Post-Marketing Surveillance – monitoring long-term safety and new uses.

MARKING SCHEME (3 Marks)

- For each correct phases named (0.5 mark)
- For correct order (1 mark).

Q.7 How do antibiotics work?

Ans. Antibiotics are chemical agents that target bacteria to either kill them (bactericidal) or inhibit their growth (bacteriostatic). They achieve this by interfering with essential bacterial processes that are absent or different in human cells. Common targets include cell wall synthesis (e.g., penicillins), protein synthesis by binding to bacterial ribosomes (e.g., tetracyclines), DNA replication (e.g., fluoroquinolones), and metabolic pathways like folic acid synthesis (e.g., sulfonamides). By disrupting these functions, antibiotics weaken or destroy the bacteria, allowing the immune system to clear the infection. Their selective action makes them effective while minimizing harm to human cells.

MARKING SCHEME (3 Marks)

- Mechanism/targets – 3 marks

Q.8 Why are monoclonal antibodies so specific?

Ans. Monoclonal antibodies (mAbs) are laboratory-produced molecules designed to bind to a single, unique epitope on a specific antigen. This specificity arises because all antibody molecules in a monoclonal preparation are identical clones derived from a single parent B-cell. The precise interaction ensures that mAbs act only on the intended target, such as a cancer cell receptor or a pathogenic protein, while leaving other cells and tissues unaffected. This high specificity reduces off-target effects and side effects compared to traditional drugs, making monoclonal antibodies particularly valuable in targeted therapy, diagnostics, and immunotherapy for diseases such as cancer, autoimmune disorders, and viral infections.

MARKING SCHEME (3 Marks)

- Reason for specificity – 3 marks



Long Answer Questions

Q.1 Explain drug discovery and development process.

Ans. Please See SLO [B-12-T-01]

Q.2 Discuss the types antibiotics and their mode of action.

Ans. Please See SLO [B-12-T-02]

Q.3 What are the advantages of monoclonal antibodies?

Ans. Please See SLO [B-12-T-05]

Q.4 Compare monoclonal antibodies to other drug classes.

Ans. Please See SLO [B-12-T-05]

